Name:_____ Period:_____

UNIT 1:

INTRODUCTION TO EARTH AND SPACE SCIENCE

ANCHOR CHARTS



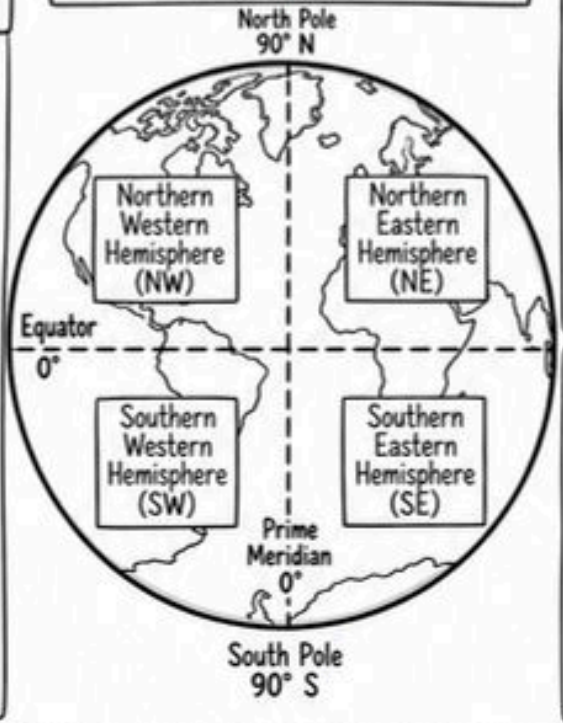
HEMISPHERES & SPHERES OF EARTH

1 What are Hemispheres?

- A hemisphere is one half of a sphere.
- Earth can be divided into Northern & Southern Hemispheres (by the Equator) and Eastern & Western Hemispheres (by the Prime Meridian).
- Each hemisphere is half of the Earth.

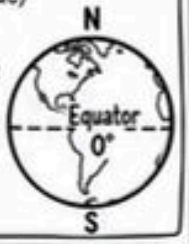
Hemisphere = 1/2 of Earth
(There are 4 hemispheres total!)

THE FOUR HEMISPHERES



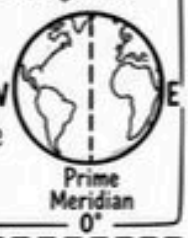
2 Hemispheres by Latitude (North & South)

- The Equator (0° latitude) divides Earth into:
  - Northern Hemisphere (0° to 90° N)
  - Southern Hemisphere (0° to 90° S)



3 Hemispheres by Longitude (East & West)

- The Prime Meridian (0° longitude) divides Earth into:
  - Eastern Hemisphere (0° to 180° E)
  - Western Hemisphere (0° to 180° W)



THE SPHERES OF EARTH

Earth is made up of interacting systems, or spheres. They all work together!

4 HYDROSPHERE

- All of Earth's water in any form.
- Includes oceans, lakes, rivers, groundwater, and water vapor.



Example:
The Atlantic Ocean, Lake Erie, rivers, groundwater, and water vapor are part of the hydrosphere.

5 ATMOSPHERE

- The layer of gases that surrounds Earth.
- Includes nitrogen, oxygen, carbon dioxide, water vapor, and other gases.



Example:
The air we breathe and the weather are part of the atmosphere.

6 GEOSPHERE (LITHOSPHERE)

- The solid parts of Earth.
- Includes rocks, minerals, soil, and Earth's landforms (crust, mantle, core).
- The lithosphere is the rigid, outer layer (crust and uppermost mantle).



Example:
Mountains, volcanoes, rocks, soil, and the Earth's crust are part of the geosphere (lithosphere).

7 BIOSPHERE

- All living things on Earth.
- Includes plants, animals, humans, and microorganisms.



Example:
Forests, oceans (living organisms), and people are part of the biosphere.

8 CRYOSPHERE

- All of Earth's frozen water.
- Includes glaciers, ice sheets, sea ice, ice caps, snow, and permafrost.



Example:
Antarctica, Arctic sea ice, glaciers, and snow on mountains are part of the cryosphere.

HEMISPHERES AND SPHERES OF EARTH

Part A: Fill in the Blank

1. A _____ is one half of a sphere.
2. The Earth can be divided into the Northern and Southern Hemispheres by the _____.
3. The Eastern and Western Hemispheres are divided by the _____.
4. The _____ includes all of Earth's water in any form, such as oceans, lakes, and groundwater.
5. The layer of gases that surrounds Earth is called the _____.

Part B: Matching

Match each sphere of Earth to its example.

Sphere	Answer	Example
1. Hydrosphere	_____	A. Antarctica, glaciers, and snow on mountains
2. Atmosphere	_____	B. The air we breathe and the weather
3. Geosphere (Lithosphere)	_____	C. Forests, oceans (living organisms), and people
4. Biosphere	_____	D. Mountains, volcanoes, rocks, and soil
5. Cryosphere	_____	E. The Atlantic Ocean, Lake Erie, and groundwater

Part C: True / False

1. _____ The Equator is used to divide Earth into the Eastern and Western Hemispheres.
2. _____ Earth is made up of interacting systems called spheres.
3. _____ The lithosphere includes only the crust of the Earth.
4. _____ All living things on Earth are part of the biosphere.

Part D: Short Answer / Application

Use what you know, along with the anchor chart, to answer the following.

1. A volcano erupts, sending ash into the sky and releasing lava that cools into new rock on the ground. Name two spheres of Earth involved in this event and explain how each one is involved.

Sphere 1: _____ Sphere 2: _____

_____

_____

_____

2. Explain how the hydrosphere and biosphere depend on each other, using an example from your own life or experience.

_____

_____

_____

3. If you are standing at 30°N, 90°W, which two hemispheres are you located in? Explain how you know.

_____

_____

CER: CLAIM • EVIDENCE • REASONING

A CER is a way to construct a scientific explanation using evidence and what you know about science.

1 What is a CER?

- A CER (Claim, Evidence, Reasoning) is a structured response to a question or phenomenon.
- It uses evidence from data, observations, or reliable sources.
- It connects the evidence to scientific principles to explain why your claim is correct.
- A strong CER shows clear thinking and uses science to support your ideas.

2 The Parts of a CER

CLAIM

- Your answer to the question.
- A clear, one-sentence statement.
- Does not include evidence or reasoning.

EVIDENCE

- Scientific data that supports your claim.
- Can include numbers, observations, graphs, tables, or information from a text or lab.
- Must be specific and relevant.

REASONING

- Explains why the evidence supports your claim.
- Uses scientific principles, concepts, or vocabulary.
- Shows the connection between the evidence and your claim.

3 Sentence Starters

CLAIM

- I claim that...
- I think...
- My conclusion is...
- Based on the information, ...

EVIDENCE

- The data shows...
- According to the graph...
- The evidence indicates...
- In the investigation, I observed...
- The text states...

REASONING

- This evidence supports my claim because...
- This happens due to...
- This is consistent with the scientific principle that...
- Therefore...

4 CER Model

CLAIM
Answer the question.

↓

EVIDENCE
Provide data to support your claim.

↓

REASONING
Explain why the evidence supports your claim using scientific ideas.

5 Example

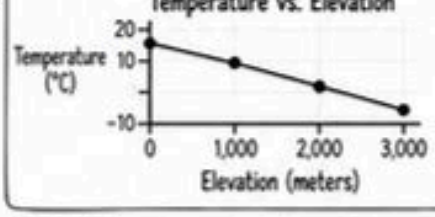
Question: Why are temperatures lower at higher elevations in the mountains?

CLAIM

I claim that temperatures are lower at higher elevations in the mountains.

EVIDENCE

The graph shows that the temperature on a mountain decreases from 15°C at sea level to -5°C at 3,000 meters.



REASONING

This evidence supports my claim because as elevation increases, air pressure decreases. Lower air pressure means the air is less dense and holds less heat, so temperatures are lower at higher elevations.

6 Tips for a Strong CER

- Answer the question directly.
- Use specific and accurate evidence (numbers, observations, or details).
- Explain your thinking using scientific concepts and vocabulary.
- Make sure all three parts are included.
- Use complete sentences.
- Stay focused on the question.
- Check that your reasoning clearly connects the evidence to your claim.



CLAIM + EVIDENCE + REASONING = A STRONG SCIENTIFIC EXPLANATION

CER: CLAIM, EVIDENCE, REASONING

Part A: Fill in the Blank

1. CER stands for Claim, _____, and Reasoning.
2. The _____ is your answer to the question and should be a clear, one-sentence statement.
3. _____ is scientific data that supports your claim, such as numbers, observations, or graphs.
4. The _____ explains why the evidence supports your claim using scientific principles or vocabulary.

Part B: Matching

Match each part of a CER to a sentence starter you could use to write it.

Part of CER	Answer	Sentence Starter
1. Claim	_____	A. "This evidence supports my claim because..."
2. Evidence	_____	B. "I claim that..."
3. Reasoning	_____	C. "The data shows..."

Part C: True / False

1. _____ A claim should include evidence to make it stronger.
2. _____ Evidence must be specific and relevant to the claim.
3. _____ Reasoning connects the evidence to the claim using scientific ideas.
4. _____ A CER response only needs a claim and evidence to be complete.

Part D: Short Answer / Application

Use what you know, along with the anchor chart, to answer the following.

1. Read the following paragraph:

"Puddles disappear faster on hot, sunny days than on cool, cloudy days. In the school parking lot, a puddle was completely gone within an hour on a sunny afternoon when it was 85°F, but a similar puddle lasted overnight when it was cloudy and only 60°F. This happens because heat causes water molecules to move faster and evaporate more quickly, so warmer temperatures speed up evaporation."

Identify the claim, the evidence, and the reasoning in this paragraph.

Claim: _____

Evidence: _____

_____

Reasoning: _____

_____

2. Write your own CER response for this question: Why does a snowball melt faster in your hand than on a table? Include a claim, evidence you would expect to observe, and reasoning that explains the science behind it.

_____

_____

_____

_____

_____

_____

READING A MAP

Maps are models of places on Earth. They give us information about location, distance, direction, and the physical and human features of an area.

1 Map Title

- Tells what the map is about.
- Often includes the area and the type of information shown.
- Always read the title first!

2 Legend (Key)

- Explains what the symbols, colors, and lines mean.
- Use the legend to identify features on the map.
- Always check the legend before reading a map!

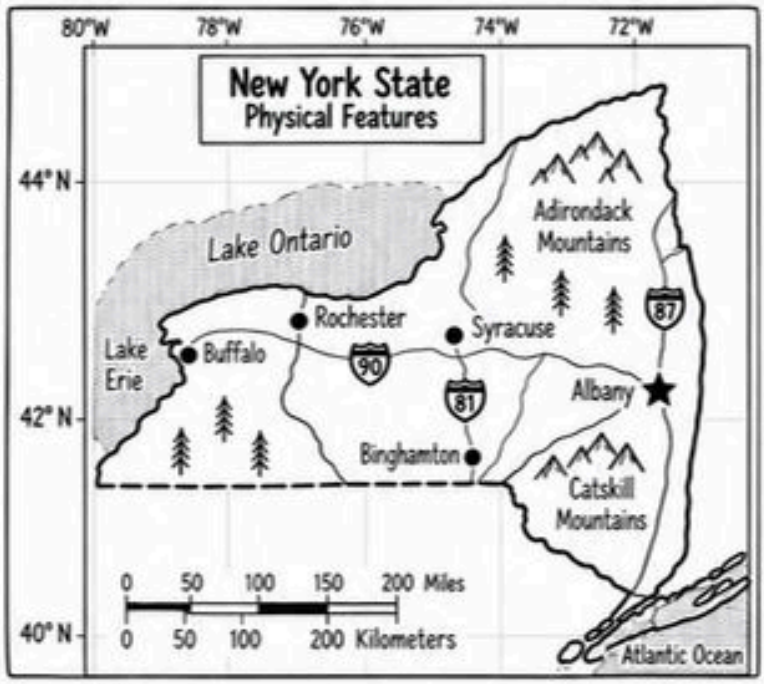
3 Compass Rose



- Shows direction on a map.
- The main directions are North (N), South (S), East (E), and West (W).
- Helps you determine the location of places relative to each other.

Legend (Key)

- ★ = State Capital
- ● = City/Town
- 90 = Interstate Highway
- = Other Road
- = River
- = Lake
- = Mountains
- = Forest
- = Higher Elevation
- = State Boundary



4 Grid / Coordinates

- The grid helps you find an exact location.
- Use latitude (north-south) and longitude (east-west) lines.
- Write latitude first, then longitude.

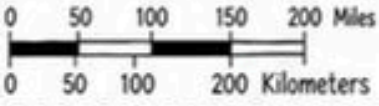
Example:
Syracuse is about 43° N, 76° W.

5 Map Scale

- Shows the relationship between a distance on the map and the actual distance on Earth.
- Can be a bar scale, ratio, or written statement.

Example (written statement):
1 cm on the map = 50 miles

Bar scale (on this map):

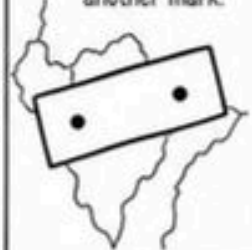


How to Use a Map Scale (with a piece of paper)

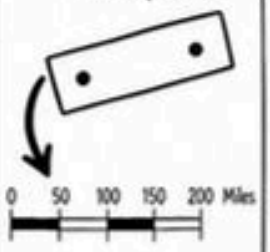
1. Place a piece of paper on the map and mark your starting point.



2. Keep the paper along the route to your ending point and make another mark.



3. Take your marked paper and drag it to the map scale.



4. See what the distance matches on the scale.

Example:
The distance from Buffalo to Syracuse is about 100 miles.

READING A MAP

Part A: Fill in the Blank

1. The _____ tells what the map is about and should always be read first.
2. The _____ explains what the symbols, colors, and lines on a map mean.
3. The _____ shows direction on a map using N, S, E, and W.
4. A map's _____ shows the relationship between distance on the map and actual distance on Earth.

Part B: Matching

Use the legend on the anchor chart to match each map symbol to its meaning.

Symbol	Answer	Meaning
1. Star (★)	_____	A. City/Town
2. Circle (●)	_____	B. River
3. Wavy line	_____	C. State Capital
4. Shaded/hatched area	_____	D. Higher elevation

Part C: True / False

1. _____ You should always check the legend before reading a map.
2. _____ A compass rose is used to measure the distance between two places.
3. _____ On the New York map, Albany is marked with a star because it is the state capital.
4. _____ Latitude lines on the map are used to measure distance east and west.

Part D: Short Answer / Application

Use the New York State Physical Features map on the anchor chart to answer the following.

1. Using the map, name one city located along Interstate 90. _____
2. Using the compass rose, is Buffalo located to the north or south of Binghamton?_____
3. Using the map scale, estimate the distance between Rochester and Albany. Explain the method you used to find your estimate.

_____ miles

_____ kilometers

Explanation:

_____

_____

LATITUDE & LONGITUDE

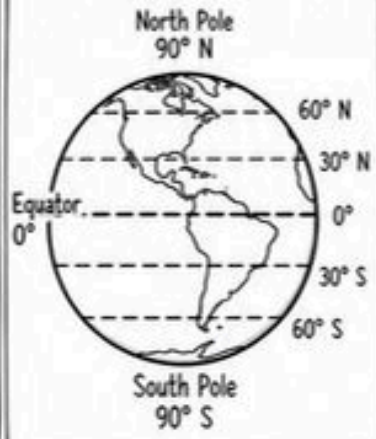
A global address system that helps us locate any place on Earth.

1 What are Latitude and Longitude?

- Latitude and longitude are imaginary lines that form a grid called a coordinate system.
- They help us find the exact location of places on Earth.
- Locations are written as latitude, longitude (e.g., 43° N, 78° W).

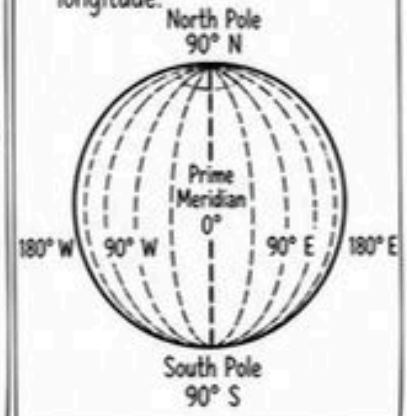
2 Latitude (North & South)

- Latitude lines run east-west (parallel to the Equator).
- Measure distance north or south of the Equator (0°).
- Range from 0° to 90° N and 0° to 90° S.
- The Equator is at 0° latitude.



3 Longitude (East & West)

- Longitude lines run north-south (from pole to pole).
- Measure distance east or west of the Prime Meridian (0°).
- Range from 0° to 180° E and 0° to 180° W.
- The Prime Meridian is at 0° longitude.



4 Hemispheres

By Latitude (North & South)

- Northern Hemisphere: 0° to 90° N
- Southern Hemisphere: 0° to 90° S

By Longitude (East & West)

- Eastern Hemisphere: 0° to 180° E
- Western Hemisphere: 0° to 180° W

The Four Hemispheres

- Northern & Eastern (NE)
- Northern & Western (NW)
- Southern & Eastern (SE)
- Southern & Western (SW)

5 Important Lines

Equator	0° latitude
Prime Meridian	0° longitude
North Pole	90° N
South Pole	90° S
International Date Line	180° longitude (roughly)

6 Finding a Location

1. Find the latitude (N or S) first.
2. Then find the longitude (E or W).
3. Write the coordinates as (latitude, longitude).

Example:
Buffalo, NY is located at about 43° N, 78° W.

7 Practice Examples

Location	Latitude	Longitude
Buffalo, NY	43° N	78° W
New York City, NY	41° N	74° W
Orlando, FL	28° N	81° W
Los Angeles, CA	34° N	118° W
London, England	51° N	0°
Tokyo, Japan	36° N	140° E
Sydney, Australia	34° S	151° E

8 Quick Tips

- **Latitude = "Ladders"** (horizontal lines)
- **Longitude = "Long"** lines (vertical lines)
- Always write latitude first, then longitude.
- Use N, S, E, W to show the hemisphere.
- Latitude and longitude work together like a global address!

LATITUDE AND LONGITUDE

Part A: Fill in the Blank

1. _____ lines run east-west and measure distance north or south of the Equator.
2. _____ lines run north-south and measure distance east or west of the Prime Meridian.
3. The Equator is located at 0° _____.
4. The Prime Meridian is located at 0° _____.
5. When writing coordinates, you should always write _____ first, then _____.

Part B: Matching

Match each line or point to its location

Line	Answer	Location
1. Equator	_____	A. 90°N
2. Prime Meridian	_____	B. 0° longitude
3. North Pole	_____	C. 180° longitude (roughly)
4. International Date Line	_____	D. 0° latitude

Part C: True / False

1. _____ Latitude lines are also called "ladders" because they run horizontally.
2. _____ Longitude lines run parallel to the Equator.
3. _____ A location with a latitude of 34°S is in the Southern Hemisphere.
4. _____ Tokyo, Japan is located in the Western Hemisphere.

Part D: Short Answer / Application

1. A city has a latitude of 5°N. Is this city closer to or farther from the Equator than Orlando, FL (28°N)? Explain your reasoning.

_____

_____

2. A location has a latitude of 15°S. Which hemisphere is it in, and how do you know?

Hemisphere: _____

Explain: _____

3. Give a set of coordinates (latitude and longitude) for a location that would be in the Southern and Western Hemispheres. Explain how you chose your answer.

Coordinates: _____

Explain: _____

_____

DENSITY

Density tells us how much mass is packed into a certain amount of volume.

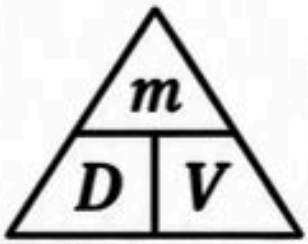
1 What is Density?

- Density is the amount of mass in a given volume.
- It tells us how tightly matter is packed together.
- More dense = more mass in the same volume.
- Less dense = less mass in the same volume.
- Density is a physical property and can be used to identify materials.

2 Formula

$$D = \frac{m}{V}$$

D = density (g/cm³ or g/mL)
m = mass (g)
V = volume (cm³ or mL)



Use this triangle to help you solve for missing values.

3 Units

- Density is usually measured in grams per cubic centimeter (g/cm³).
- 1 cm³ is equal to 1 mL, so you may also see g/mL.
- Always include the correct units in your answer.
- Water has a density of 1.00 g/cm³ (1.00 g/mL). Use this as a reference when deciding if something will float or sink.

4 How to Calculate Density

1. Measure the mass (g).
2. Measure the volume (cm³ or mL).
3. Use the formula $D = \frac{m}{V}$.
4. Include units in your answer.

Example:

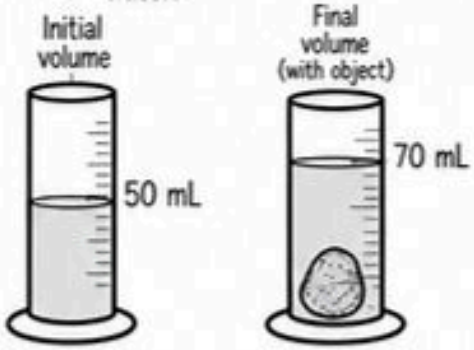
A rock has a mass of 54 g and a volume of 20 cm³. What is its density?

$$D = \frac{54\ g}{20\ cm^3} = 2.7\ g/cm^3$$

5 Water Displacement

(to find the volume of an irregular object)

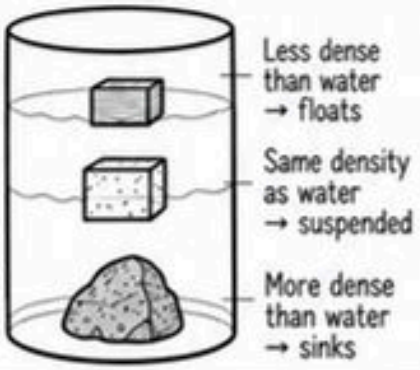
1. Fill a graduated cylinder with water. Record the initial volume.
2. Carefully place the object in the water.
3. Record the final volume.
4. Subtract to find the volume object.



Volume of object = final volume − initial volume
70 mL − 50 mL = 20 mL
So, the volume of the object is 20 mL (20 cm³).

6 Floating and Sinking

- An object will float if its density is less than the fluid it is in.
- An object will sink if its density is greater than the fluid it is in.
- If an object has the same density as the fluid, it will be neutrally buoyant (stay suspended).
- In water (1.00 g/cm³), objects with a density less than 1 float and objects with a density greater than 1 sink.



7 Key Takeaways

Density = mass ÷ volume.

- Use the correct units (g/cm³ or g/mL).
- Water has a density of 1.00 g/cm³ (1.00 g/mL).
- More dense means more mass in the same volume.
- Less dense means less mass in the same volume.
- Density helps us identify materials.
- Water displacement can be used to find the volume of an irregularly shaped object.

DENSITY

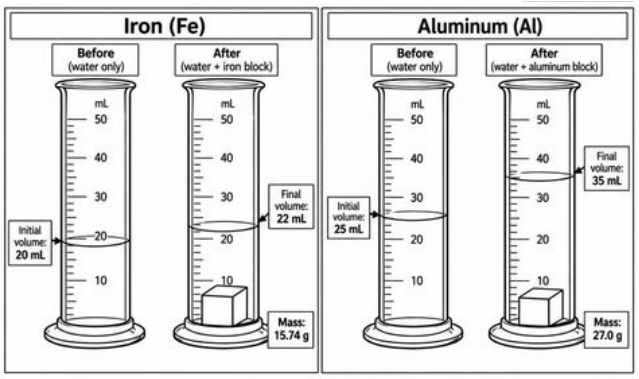
Part A: Fill in the Blank

1. Density is the amount of _____ in a given _____.
2. The formula for density is D = _____ ÷ _____.
3. Density is usually measured in grams per cubic centimeter, written as _____.
4. An object will float if it is _____ dense than the fluid it is in.

Part B: Matching

Use the anchor chart for water. Calculate the density of iron and aluminum using the cylinder diagrams, then use reasoning to place air.

Material	Answer	Density
1. Water	_____	A. 7.87 g/cm³
2. Iron	_____	B. 1.00 g/cm³
3. Aluminum	_____	C. 2.70 g/cm³
4. Air	_____	D. 0.0012 g/cm³



Part C: True / False

1. _____ Density is a physical property that can be used to identify materials.
2. _____ If two objects have the same volume, the object with more mass is more dense.
3. _____ An object will sink if it is less dense than the fluid it is in.
4. _____ Density and mass are the same thing.

Part D: Short Answer / Application

1. A student measures a mineral sample and finds it has a mass of 40 g and a volume of 16 cm³. Calculate the density of the mineral. Show your work and include units.

2. If you placed a block of aluminum (density 2.70 g/cm³) into a container of vegetable oil (density about 0.92 g/cm³), would the aluminum float or sink? Explain your reasoning using the density values.

3. Block A has a mass of 50 g and a volume of 10 cm³. Block B has a mass of 30 g and the same volume, 10 cm³. Calculate the density of each block and determine which one is more dense.

4. A liquid has a density of 1.05 g/cm³. Would an ice cube (density 0.92 g/cm³) float or sink in this liquid? Explain your reasoning.

FEEDBACK LOOPS

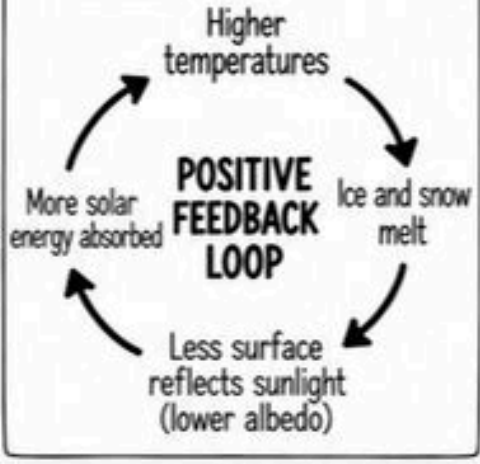
A feedback loop is a cycle in which a change in a system leads to a response that either increases (positive) or decreases (negative) the original change.

1 What is a Feedback Loop?

- A feedback loop is a series of events where an initial change leads to a response that feeds back into the system.
- Feedback loops can either increase change (positive) or decrease change (negative).
- They help explain how Earth's systems stay balanced or undergo rapid change.
- Feedback loops occur in the atmosphere, hydrosphere, geosphere, biosphere, and human systems.

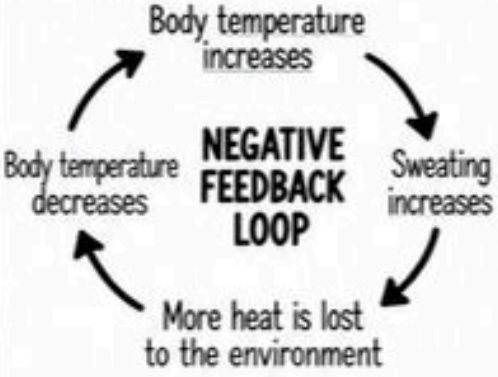
2 Positive Feedback Loop

- A positive feedback loop amplifies the initial change.
- The response makes the original change even larger.
- Can lead to rapid change.
- **Example:** Melting ice and albedo.



3 Negative Feedback Loop

- A negative feedback loop reduces the initial change.
- The response works in the opposite direction to counteract the change.
- Helps keep systems stable.
- **Example:** Body temperature regulation.



4 More Examples

Positive Feedback (increases change)	Negative Feedback (decreases change)
Melting ice → lower albedo → more warming	Increased CO_2 → more plant growth → more CO_2 removed from atmosphere
Water vapor → stronger greenhouse effect → more warming	Rising temperature → more evaporation → cooling effect
Permafrost thaw → more methane released → more warming	Carbonate rocks weathering → removes CO_2 → less warming
Forest fires → more CO_2 released → more warming	Predator population increases → prey population decreases → stabilizes ecosystem

5 How to Identify

- Look at the direction of the response.
- If the response makes the initial change bigger → positive feedback.
- If the response works against the initial change → negative feedback.
- Think about the overall effect on the system.

6 Key Takeaways

- Positive feedback loops amplify change and can lead to rapid shifts in Earth's systems.
- Negative feedback loops reduce change and help maintain stability.
- Feedback loops are important for understanding natural processes and human impacts.
- Both types of feedback loops occur in Earth's interconnected systems.

FEEDBACK LOOPS

Part A: Fill in the Blank

1. A _____ is a series of events where an initial change leads to a response that feeds back into the system.
2. A positive feedback loop _____ the initial change.
3. A negative feedback loop _____ the initial change.
4. Feedback loops help explain how Earth's systems stay _____ or undergo rapid change.
5. An example of a negative feedback loop on the anchor chart is _____ regulation.

Part B: Matching

Match each term to its definition.

Term	Answer	Definition
1. Feedback Loop	_____	A. A series of events where an initial change leads to a response that feeds back into the system.
2. Positive Feedback Loop	_____	B. Works in the opposite direction to counteract the change and helps keep systems stable.
3. Negative Feedback Loop	_____	C. Amplifies the initial change and can lead to rapid change.

Identify the type of Feedback Loop for each scenario:

1. Melting ice and albedo: (circle) Negative feedback loop Positive feedback loop
2. Body temperature regulation: (circle) Negative feedback loop Positive feedback loop

Part C: True / False

1. _____ A positive feedback loop reduces the initial change.
2. _____ Negative feedback loops help keep systems stable.
3. _____ Melting ice and albedo is an example of a negative feedback loop.
4. _____ Feedback loops are found only in the atmosphere.
5. _____ If a response works against the initial change, it is a negative feedback loop.

Part D: Short Answer / Application

1. Excess fertilizer runs off into a lake, causing algae to grow rapidly. When the algae die, they are decomposed by bacteria that use up oxygen in the water. This drop in oxygen kills fish and other organisms, which are then decomposed too, using up even more oxygen. Is this an example of a positive or negative feedback loop? Explain your reasoning using the definitions on the anchor chart.

   Type of feedback loop:_____

   _____

   _____

2. Describe a real or hypothetical example of a feedback loop (positive or negative) that is different from the ones on your anchor chart. Identify the initial change and the response, then explain whether it is a positive or negative feedback loop and why.

   Type of feedback loop:_____

   _____

   _____